

Sai Pushkar Sikharam

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SUMMARY

Software Engineer and M.S. Computer Science candidate with hands-on experience building backend services, RESTful APIs, and full-stack web applications. Proficient in Python, C++, JavaScript, TypeScript, Next.js, React, FastAPI, PostgreSQL, Docker, and AWS. Experienced in designing data-driven applications, integrating AI services, and building reliable, maintainable systems with a focus on clean API design and practical user workflows.

EDUCATION

Texas A&M University – Corpus Christi
Master's Computer Science

Aug 2024 – May 2026

- **Coursework:** Machine Learning, Deep Learning, Advanced Software Engineering, DSA, Transformers, ML Ops, LLM

SCSVMV University – Kanchipuram

May 2020 – May 2024

Computer Science and Engineering

- **Coursework:** Operating Systems, Computer Networks, DBMS, OOPs, Compiler Design, Web Technologies, Cloud Computing

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, TypeScript, SQL
- **Frontend:** React.js, Next.js, HTML, CSS, Tailwind CSS
- **Backend & APIs:** FastAPI, Flask, REST APIs, Authentication, JWT, Pydantic
- **Databases:** SQLite, PostgreSQL, MongoDB
- **Cloud:** AWS, GCP, Docker, CI/CD, GitHub
- **AI Integration:** Gemini AI, OpenAI API, Hugging Face, LLM Integration, NLP, RAG
- **Concepts:** OOP, Data Structures & Algorithms, API Design, Database Migrations, Microservices

WORK EXPERIENCE

Texas A&M University - Corpus Christi
Graduate Research Assistant

Jun 2025 – May 2026

- Architected an automated GitHub CVE/CWE security auditing pipeline analyzing 89 open-source repositories, correlating Issues, PRs, Commits, and Releases into unified findings.
- Implemented deterministic vulnerability detection (regex + confidence scoring), fix-version resolution (SHA-to-tag mapping), and rate-limit-aware scalable API orchestration.
- Built *Crypto Fusion*, FastAPI-based backend services exposing ML inference pipelines as production-ready APIs, integrating caching, structured logging, and robust error handling to improve reliability and reduce failed requests by 30% under high-load testing.
- Designed high-throughput data ingestion pipeline processing thousands of daily financial and new records, integrating LLM-based summarization and sentiment services via asynchronous batching reducing model load by 40% & cutting inference to 5s.

Math Buddy

Python Developer

Mar 2022 – Dec 2022

- Built a Python-based desktop voice assistant using speech recognition and text-to-speech libraries for real-time audio input processing and command execution.
- Designed test cases and hands-on programming assignments covering OOP, algorithms, debugging, and output validation.

PROJECTS

Kinetic – Adaptive Nutrition Platform | Next.js 15, FastAPI, PostgreSQL, Docker

Jan 2026 – May 2026

- Built a full-stack nutrition tracking platform with JWT authentication, 8+ REST API modules, AI-assisted chat, and a Docker Compose-based development environment using Next.js 15, FastAPI, and SQL-backed data models.
- Designed an adaptive calorie engine using EWMA smoothing and regression analysis to generate behaviour-based recommendations from user tracking data.

Talk2Campus - Fast API, ReactJS, Python, Mongo DB, OpenAI API

Aug 2025 – Dec 2025

- Built an AI campus assistant with REST APIs for authentication, query orchestration, and document retrieval, integrating a RAG pipeline with LLM processing for low-latency campus question answering.

ACHIEVEMENTS

- *Machine Learning-Based Analysis and Forecasting of Cryptocurrency Trends, " ICCSMM, 2025*
- **IEEE 2023: Published research paper titled "Helmet Detection and Number Plate Recognition using YOLOv3"**